

USER GUIDE

How to design, test and publish customer documents with the Document Designer: building pages from blocks, binding them to message data, previewing with test data, sharing clauses, importing existing Word, PDF, Documaker and GhostDraft documents, and publishing versions safely.

Who it is for: Document authors, business analysts and reviewers. No programming is needed; a few advanced notes mention the Liquid template language for people who are curious about what the designer produces.

Edition: Working draft (proof of concept) · Updated 10/01/2026

CONTENTS

1. Getting started	What the designer does and a tour of the screen.
2. Building a document	Blocks, text, layout and styling.
3. Working with data	The data model, fields, formats, lists and conditions.
4. Test data and previews	Preview PDF, test scenarios and Preview all.
5. Versions and publishing	Drafts, publishing safely, comparing and rolling back.
6. Shared clauses	Write wording once and use it in many documents.
7. Importing existing documents	Word, PDF, Documaker and GhostDraft.
8. Finding where fields are used	Which templates and clauses use a value.
9. Reference	Limits and advanced notes.

CHAPTER 1: GETTING STARTED

The Document Designer is where customer documents are laid out. A document template describes the page: text, headings, tables, logos and the places where policy data is printed. When a policy event happens, the document service fills the published template with that policy's data and produces the PDF.

WHAT YOU CAN DO

Design documents visually by dragging blocks onto the page and typing directly on it.

Bind parts of the page to message data (policy number, insured name, locations, claims) so every generated document shows the right values.

Preview the result as a PDF with example data or with any number of saved test scenarios.

Keep versions: work on a draft while the published version keeps generating documents, review what changes before publishing, and roll back if needed.

Reuse wording across documents with shared clauses, and bring existing Word, PDF, Documaker and GhostDraft documents into the designer.

THE SCREEN AT A GLANCE

The top bar holds the main actions: Publish, Save Draft and Preview PDF. The three-dot menu next to them has Show Liquid, Compare with Published, Render Published, the import commands and Discard Draft.

The toolbar underneath chooses what you are working on: Kind (Template or Clause), the name, the saved Version with its Open button and a badge showing the open version and its status. On the right, Test data picks the data used for sample values and previews, Scenarios manages test data, and Data fields show switches how bound fields look on the canvas.

The left panel has the Model (the data your document can print) and the Blocks you can drag onto the page. The page itself is in the middle. The right panel has three tabs: Settings for the selected element, Styles for its look, and Layers for the structure of the page.

Tip: Hover over any button to see what it does.

OPENING A DOCUMENT

Every template has a name made of letters, numbers, dashes and underscores (for example cpp-declarations). The designer opens the newest saved version of the name shown in the toolbar when it starts.

How to

1. Choose Template in Kind (or Clause for a shared clause).
2. Type the name and press Tab. The Version list fills with the saved versions.
3. Pick a version and click Open.
4. To start a new document, type a new name: the status bar says it is new, and Save Draft creates version 1.

Tip: If you have unsaved changes the designer asks before replacing them.

CHAPTER 2: BUILDING A DOCUMENT

Documents are built from blocks. Drag a block from the Blocks panel onto the page, then type into it or change it in the Settings and Styles tabs.

BLOCKS

Brand blocks give the MOE look: Document Header (logo and title), MOE Logo, Card, Callout and Lead-in.

Basic blocks are Heading, Subheading and Text.

Data blocks print message data: Data Table, Repeat, Show If, Data Image and Clause. Single values are added from the Model panel instead (see Working with data).

Layout blocks arrange the page: 2 Columns, 3 Columns, 1/3 + 2/3, 2/3 + 1/3, Divider and Page Break. Material blocks add icons, banners and cards. Form Page blocks build fixed-position form pages like converted Documaker forms.

How to

1. Find the block in the Blocks panel.
2. Drag it onto the page. A green line shows where it will land.
3. Click the new block to select it; its options appear in the Settings tab.

Tip: Select an element and press Delete to remove it, or use the toolbar that appears above the selection to move, copy or delete it.

COLUMNS AND PAGE BREAKS

Column blocks place content side by side. Select a column to set its width (1/4, 1/3, 1/2 and so on) or drag its edge; select the row to set the gap between columns and how they line up. Columns still flow and paginate in the PDF.

A Page Break starts a new page in the PDF. On the canvas it shows as a dashed red line labelled page break.

STYLING

Select an element and open the Styles tab to change fonts, colors, spacing, borders and alignment. Styles apply to the selected element; classes shown at the top of the tab let several elements share a style.

The Layers tab shows the structure of the page as a tree. Use it to select elements that are hard to click on the page, or to drag them into a different place. Invisible helper elements that the designer adds to protect imported text are kept out of the Layers tree.

Tip: Show Liquid in the three-dot menu shows exactly what the designer will save: the page as HTML with Liquid placeholders, and its CSS.

CHAPTER 3: WORKING WITH DATA

A template prints values from the message the document service receives. The Model panel shows the shape of that message: groups of values, single values and lists.

THE DATA MODEL

The model is an example message in JSON. Its shape builds the Model tree; its values are what the canvas and Preview PDF show. The model is saved with each template version.

Lists (shown with square brackets) become Repeats and Data Tables. Inside a list, each item has a short name: in a list called locations each item is location, so a location's name is location.name.

How to

1. To change the model, click the edit button in the Model panel, change the JSON and click Apply.
2. To load a model from a file, click the upload button and choose a JSON example message or a JSON Schema. For a schema the designer makes up example values.

Tip: Property names must be letters, numbers and underscores to be used in a template. Others are shown greyed out.

Note: Models are limited to 2 MB.

ADDING DATA FIELDS

A data field prints one value from the message, such as the policy number. On the canvas it is tinted so you can tell it apart from fixed text.

How to

1. Drag a value from the Model panel onto the page.
2. To rebind a field to a different value, select the field on the page and click the value in the Model panel.
3. In the Settings tab choose a Format and, optionally, what to print when the value is empty (If empty, show).

Tip: Dragging a group (for example policy) adds a label and a field for each of its values in one go.

FORMATS

Formats control how a value prints: Currency, Whole dollars (\$1,234), Percent, Number (1,234), Decimal (1,234.00), Date (MM/dd/yyyy) and UPPERCASE. Without a format the value prints as it is in the message.

If empty, show prints a replacement such as Not Applicable when the value is missing or blank.

LISTS: REPEAT AND DATA TABLE

A Repeat prints its contents once for every item of a list: one card per location, one line per claim. A Data Table prints a table with one row per item and can add a totals row.

Dragging a list from the Model panel onto the page asks whether you want a table, cards or a list, and which of the item's values to show.

A Repeat can show a message when the list is empty (If empty, show), and can print only the items that meet a condition (Only if).

Tip: On the canvas, Repeats are outlined in purple and labelled with the list they repeat over.

SHOW IF

Show If prints its contents only when a condition is met, for example only for renewals or only when there are claims. Conditions are: has a value / is not empty, is empty, equals, does not equal, is greater than, is less than and contains.

How to

1. Drag Show If from the Data blocks onto the page and put the conditional content inside it.
2. With Show If selected, click the value to test in the Model panel.
3. Choose the Condition and, for comparisons, type the Value.

DATA IMAGES

A Data Image prints an image that arrives in the message, for example a vendor map or photo. Bind it to the image value like any other field.

HOW FIELDS LOOK ON THE CANVAS

Data fields show switches the canvas between Sample values (what the PDF will print with the selected test data), Field names, and Liquid (the exact placeholder that is saved). The saved template is the same in every mode; hover a field to see its full expression.

BINDING PROBLEMS

When a field, list or condition points at something the model doesn't have, it is outlined in red and listed under the Model panel. Click a problem to select the element, then click the right value in the Model panel to fix it.

Fields inside a list only work inside a Repeat or Data Table over that list; the problem list says so when one is placed outside.

CHAPTER 4: TEST DATA AND PREVIEWS

Always look at the document the way customers will see it. The designer renders the PDF with the same engine the document service uses.

PREVIEW PDF

Preview PDF renders what is on the canvas right now (saved or not) with the data chosen in Test data. The PDF opens in a window with links to open it in a new tab or download it.

TEST DATA SCENARIOS

One example is rarely enough. Scenarios are named sets of test data, such as Minimal policy, Many locations or No claims, saved for the template name. They are kept across versions and are never published.

Pick a scenario in Test data to see its values on the canvas and in Preview PDF. The status bar lists any fields that scenario has no value for, and any lists it leaves empty.

How to

1. Click Scenarios in the toolbar.
2. Click New scenario. The editor starts from the data currently shown.
3. Give it a name, change the JSON and click Save scenario.
4. Use Show to switch to a scenario, Edit to change or rename it, and Delete to remove it.

Tip: Make a scenario for each edge case you care about: missing optional data, long names, many list items, zero amounts.

Note: Up to 50 scenarios per template, 2 MB each.

PREVIEW ALL

Preview all renders the model example and every scenario in one go. The list on the left shows each one's page count, empty fields or the reason it failed; click one to see its PDF.

How to

1. Click Scenarios, then Preview all.
2. Click each result to check its PDF.

Note: Preview all renders up to 20 scenarios at a time.

CHAPTER 5: VERSIONS AND PUBLISHING

Every template keeps its history. Customers only ever get the Published version; you work on a Draft until it is ready.

DRAFT, PUBLISHED AND RETIRED

Draft is work in progress. Save Draft saves the canvas to the draft (or starts a new draft when the open version is Published or Retired, which are read-only).

Published is the live version used for generated documents. There is only one at a time.

Retired versions were published before. They are kept so you can look at them or roll back.

Tip: The badge next to the Version list shows the open version and its status, followed by • unsaved when there are unsaved changes.

PUBLISHING AND REVIEWING CHANGES

Publish saves the canvas as a draft if needed and makes it the live version; the previous published version is retired.

When a version is already published, the designer first shows Review changes before publishing: the printed text with additions and removals marked, both PDFs side by side, and the HTML and CSS differences. Both versions are rendered with the same test data so only template changes show up. Click Publish to go ahead or Cancel to keep the current version live.

How to

1. Click Publish.
2. Read the summary and the Printed text tab; check PDFs side by side for layout changes.
3. Click Publish vN, or Cancel.

Tip: Compare with Published in the three-dot menu shows the same review at any time without publishing.

ROLLING BACK

To go back to an earlier version, open the retired version and click Publish. The review shows what will change, and the button reads Roll back to vN.

DISCARD DRAFT AND RENDER PUBLISHED

Discard Draft (three-dot menu) deletes the draft version. It cannot be undone.

Render Published renders the published version on the server exactly as the document service would, which is useful to confirm what customers are currently receiving.

CHAPTER 6: SHARED CLAUSES

A clause is a reusable piece of a document, such as a standard exclusion, a disclaimer or a contact block. Clauses have their own versions; publishing a new version of a clause updates every template that uses it.

CREATING A CLAUSE

Clauses are designed on the same canvas as templates and can use data fields, lists and conditions. They use the data of the document they are placed in.

How to

1. Set Kind to Clause and type the clause name.
2. Design the clause on the canvas.
3. Save Draft, then Publish when it is ready. Only published clauses can be used in templates.

Note: A clause can't include itself, directly or through another clause.

USING A CLAUSE IN A TEMPLATE

The Clause block places a clause in the template. The canvas shows the clause's current wording (read-only) with a teal outline. By default the latest published version prints, so the template picks up clause updates automatically. Pin a version to keep a specific wording.

If a clause is missing or not yet published, the canvas and Preview PDF say so.

How to

1. Drag the Clause block from the Data blocks onto the page.
2. In the Settings tab choose the clause.
3. Optionally type a version number in Pin version.

Tip: To change the wording, switch Kind to Clause, open the clause and publish a new version. Templates that pin an older version are not affected.

CHAPTER 7: IMPORTING EXISTING DOCUMENTS

Start from an existing document instead of a blank page. Imports open on the canvas as a new, unsaved template named after the file.

WORD DOCUMENTS (.DOCX)

Word documents become flowing content: headings, paragraphs, lists, tables, images, links and page breaks. The Word page margins are kept, and the default header and footer become a running header and footer with page numbers (Page X of Y).

Mail-merge fields and content controls become data fields, and a data model is built from them. Placeholder text such as {{ policy.number }}, «Insured Name» and, optionally, [Insured Name] can also become data fields; the import dialog lets you choose which kinds to look for. Word field switches (uppercase, date and number pictures) become field formats.

How to

1. Three-dot menu, Import PDF / Word, and choose the .docx file.
2. Tick the kinds of placeholder text to turn into data fields and click Import.
3. Read the status bar: it lists what was imported and any notes, such as first-page or multi-section headers that were not carried over.

Note: Old .doc files must be saved as .docx in Word first.

PDF DOCUMENTS

PDF pages are imported as fixed form pages: text, lines, boxes, images and form fields keep their positions. Fonts embedded in the PDF are used where they can be, so spacing matches the original.

Fillable PDF form fields are kept as form fields to be mapped to the model (see Mapping form fields).

PDFs longer than 100 pages can't be imported in one go; the designer asks which pages to import (From and To).

How to

1. Three-dot menu, Import PDF / Word, and choose the PDF.
2. For a long PDF, enter the page range and click Import.

Note: Scanned PDFs (pictures of pages) have no text to import yet; text recognition is planned.

MAPPING FORM FIELDS

Imported form fields start unmapped. The Model panel counts them and can suggest a model value for each by comparing names (it understands abbreviations such as NO, EFF and INSD, spelling slips, and whether a field holds a date or an amount).

How to

1. Click Suggest mappings under the Model panel.
2. Check the suggestions. Confident matches are already ticked; change or untick others.
3. Click Apply. Mapped fields get a sensible format for their type.
4. Map any remaining fields by selecting them and clicking a value in the Model panel.

DOCU MAKER AND GHOSTDRAFT FORMS

Import Legacy Form opens a Documaker form converted to HTML by the fact-pdf-tools converter. Import GhostDraft Form opens a GhostDraft template converted by the gd2designer tool, including its fill points, conditions, lists and a sample model.

CHECKING IMPORT QUALITY

The team checks the importer against real documents by rendering each imported document and comparing it with the original, both by words and visually. Documents that score below 90% are sent to the import review page, which shows the original, the rendering and colored overlays (original red, rendering blue, both black) so differences are easy to spot and record.

CHAPTER 8: FINDING WHERE FIELDS ARE USED

Before a message field is renamed or retired, find every document that prints it.

FIELD USAGE SEARCH

The search looks through every template and clause (their newest and published versions) and lists the ones that use the field, how often, and through which clauses. A parent finds everything under it: policy also finds policy.number. Fields inside lists are written with [] (locations[].name). The current, unsaved canvas is counted too.

How to

1. Click the filter button in the Model panel heading, or hover a value in the Model tree and click its filter button.
2. Type or pick the field and click Search. Tick All versions to include retired versions.
3. Click Open on a result to load that version into the designer.

CHAPTER 9: REFERENCE

Details that are useful to know once you are comfortable with the designer.

LIMITS

Data model: 2 MB. Test scenarios: 50 per template, 2 MB each; Preview all renders up to 20. PDF import: 100 pages at a time. Names: letters, numbers, dashes and underscores, up to 64 characters.

LIQUID, FOR THE CURIOUS

The designer saves templates as HTML with Liquid placeholders, the format the document service fills in. A data field is saved as `{{ policy.number }}`, a Repeat as `{% for location in locations %} ... {% endfor %}`, Show If as `{% if ... %}` and a Clause as `{% include 'name' %}`. Show Liquid in the three-dot menu shows the whole template.

Values from the message are always printed as text, so message data can never change the layout of the page.